- Mutations affecting single base or base pair of DNA are called point mutation
- Mutations altering the number of copies of a small repeated nucleotide sequence within a gene

**Point mutation**

It refers to alterations of single base pairs of DNA or of a small number of adjacent base pairs

**Types of point mutations**

Point mutation in DNA are categorised into two main types. They are base pair

| S.No | Basis of classification | Major types of mutations | Major features |
|---|---|---|---|
| 1. | Origin | Spontaneous | Occurs in the absence of known mutagen |
| | | Induced | Occurs in the presence of known mutagen |
| 2. | Cell type | Somatic | Occurs in non-reproductive cells |
| | | Germ-line | Occurs in reproductive cells |
| 3. | Effect on function | Loss-of-function (knockout, null) | Eliminates normal function |
| | | Hypomorphic(leaky) | Reduces normal function |
| | | Hypermorphic | Increases normal function |
| | | Gain-of-function (ectopic expression) | Expressed at incorrect time or inappropriate cells |
| 4. | Molecular change | **Nucleotide substitution**<br>• Transition | A base pair in DNA duplex is replaced with a different base pair Purine to purine(A⟶G)or pyrimidine to pyrimidine(T⟶C) |
| | | • Transversion | Purine to pyrimidine(A⟶T) or pyrimidine to purine(C⟶G) |
| | | • Insertion | One or more extra nucleotides are present |
| | | • Deletion | One or more nucleotides are missing |
| 5. | Effect on translation | • Silent (synonymous) | No change in amino acid encoded |
| | | • Missense (non-synonymous) | Change in amino acid encoded |
| | | • Nonsense(termination) | Creates translational termination codon (UAA, UAG, or UGA) |
| | | • Frameshift | Shifts triplet reading of codons out of correct phase |

**Table 3.7:** Major types of mutations



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